Probability Theory and Statistics

Exercise 5 solutions

10.13. – 10.17.

Discrete Joint Distributions, Independence, Introduction to Continuous Distributions

1. a) They are not independent. A correct intuition leading to this is, for example, that if we roll a six, then we also roll an even number; hence, knowing the number of sixes affects the behavior of the number of evens (and vice versa). For instance, the number of even outcomes is always at least the number of sixes.
   To prove non-independence precisely, it suffices to find a pair of values $k \in \mathrm{Ran}(X)$ and $l \in \mathrm{Ran}(Y)$ with $\mathbb{P}(X = k, Y = l) \neq \mathbb{P}(X = k)\mathbb{P}(Y = l)$. Here $\mathrm{Ran}(X) = \mathrm{Ran}(Y) = \{0, 1, 2\}$, since among two rolls we can have $0, 1$, or 2 sixes/evens. Take $k = 2$ and $l = 1$. Then $\mathbb{P}(X = 2, Y = 1) = \mathbb{P}(2 \text{ sixes out of 2 rolls and exactly 1 even}) = 0$, while $\mathbb{P}(X = 2)\mathbb{P}(Y = 1) \neq 0$, because $\mathbb{P}(X = 2) = \frac{1}{36}$ and $\mathbb{P}(Y = 1) = 2 \cdot \frac{3}{6} \cdot \frac{3}{6}$ (since $Y = 1$ precisely when the first roll is even and the second odd, or vice versa).
   b) The full joint distribution table is obtained similarly. The events $\{X = 1, Y = 0\}$ and $\{X = 2, Y = 0\}$ are impossible. We have $\mathbb{P}(X = Y = 0) = \mathbb{P}(\text{both rolls odd}) = \frac{3}{6} \cdot \frac{3}{6} = \frac{9}{36}$, $\mathbb{P}(X = 0, Y = 1) = \mathbb{P}(\text{one roll odd, the other is 2 or 4}) = 2 \cdot \frac{3}{6} \cdot \frac{2}{6} = \frac{12}{36}$, $\mathbb{P}(X = Y = 1) = \mathbb{P}(\text{one roll odd, the other is 6}) = 2 \cdot \frac{3}{6} \cdot \frac{1}{6} = \frac{6}{36}$, $\mathbb{P}(X = 0, Y = 2) = \mathbb{P}(\text{both rolls are 2 or 4}) = \frac{2}{6} \cdot \frac{2}{6} = \frac{4}{36}$, $\mathbb{P}(X = 1, Y = 1) = \mathbb{P}(\text{one roll odd, the other is 6}) = 2 \cdot \frac{1}{6} \cdot \frac{3}{6} = \frac{6}{36}$, $\mathbb{P}(X = 1, Y = 2) = \mathbb{P}(\text{one roll is 6, the other is 2 or 4}) = 2 \cdot \frac{1}{6} \cdot \frac{2}{6} = \frac{4}{36}$, $\mathbb{P}(X = Y = 2) = \mathbb{P}(\text{both rolls are 6}) = \frac{1}{36}$.
   In table form (as usual, the marginals of X are obtained by summing columns, and those of Y by summing rows):

$Y \backslash X$	0	1	2	p_Y
0	$\frac{9}{36}$	0	0	$\frac{9}{36}$
1	$\frac{12}{36}$	$\frac{6}{36}$	0	$\frac{18}{36}$
2	$\frac{4}{36}$	$\frac{4}{36}$	$\frac{1}{36}$	$\frac{9}{36}$
p_X	$\frac{25}{36}$	$\frac{10}{36}$	$\frac{1}{36}$	1

   Note: The marginal of X is binomial with parameters $n = 2$, $p = 1/6$, and the marginal of Y is binomial with $n = 2$, $p = 1/2$.
2. a) $\dfrac{1}{60}$, because the entries in the joint distribution table must sum to 1, and this sum equals $60p$.
   b) $\mathbb{P}(X \leq 0, Y = 1) = \mathbb{P}(X = -1, Y = 1) + \mathbb{P}(X = 0, Y = 1) = 5p + 15p = \dfrac{1}{3}$.
   c) Yes. Compute the marginals. For Y, the row sums give: $\mathbb{P}(Y = -1) = p + 3p + 6p = \dfrac{1}{6}$, hence $\mathbb{P}(Y = 1) = \dfrac{5}{6}$. For X, the column sums give: $\mathbb{P}(X = -1) = p + 5p = \dfrac{1}{10}$, $\mathbb{P}(X =$

$0) = 3p + 15p = \dfrac{3}{10}$, hence $\mathbb{P}(X = 1) = \dfrac{6}{10}$. It is then easy to check that $\mathbb{P}(X = k, Y = l) = \mathbb{P}(X = k)\mathbb{P}(Y = l)$ holds for all $k \in \{-1, 0, 1\}$ and $l \in \{-1, 1\}$.

(We omit the arithmetic here, but to prove independence you must verify this for every *such (k, l), not just one!)*

d) We can compute $\mathbb{E}(XY)$ in two (equivalent) ways.

Method 1: find the distribution of XY and use the usual definition of expectation. $\mathrm{Ran}(XY) = \{-1, 0, 1\}$. $\mathbb{P}(XY = 0) = \mathbb{P}(X = 0 \text{ or } Y = 0) = \mathbb{P}(X = 0) = \dfrac{3}{10}$, $\mathbb{P}(XY = -1) = \mathbb{P}(X = 1, Y = -1) + \mathbb{P}(X = -1, Y = 1) = \dfrac{11}{60}$, hence $\mathbb{P}(XY = 1) = 1 - \frac{3}{10} - \frac{11}{60} = \frac{31}{60}$. Therefore

$$\mathbb{E}(XY) = (-1) \cdot \mathbb{P}(XY = -1) + 0 \cdot \mathbb{P}(XY = 0) + 1 \cdot \mathbb{P}(XY = 1) = \frac{31 - 11}{60} = \frac{1}{3}.$$

Method 2: use the formula for transformed expectations in two dimensions ($XY = g(X, Y)$ with $g(x, y) = xy$):

$$\mathbb{E}(XY) = \sum_{k \in \mathrm{Ran}(X)} \sum_{l \in \mathrm{Ran}(Y)} kl\, \mathbb{P}(X = k, Y = l) = p - 5p - 6p + 30p = 20p = \frac{1}{3}.$$

3. The joint distribution is:

Y \ X	1	2	3	4	5
0	1/3	2/15	1/30	0	0
1	0	2/15	2/15	1/15	0
2	0	0	1/30	1/15	1/15

Explanation: The number of draws until stopping equals the index of the first red, since we stop as soon as a red is drawn.

If a red is drawn first, clearly no white is drawn before it; hence $\mathbb{P}(X = 1, Y = 1) = \mathbb{P}(X = 1, Y = 2) = 0$. Similarly, if the red is second, we cannot have drawn two whites before it, so $\mathbb{P}(X = 2, Y = 2) = 0$. If the red is fourth, we must have drawn at least one white before it, because besides red and white there are only 2 green balls; hence $\mathbb{P}(X = 4, Y = 0) = 0$. If the red is fifth, both whites must already have been drawn; thus $\mathbb{P}(X = 5, Y = 0) = \mathbb{P}(X = 5, Y = 1) = 0$.

The probability that the first draw is red is $1/3$, since one third of the balls are red.

The event $\{X = 2, Y = 0\}$ means: red is drawn on the second draw, and the first draw is not white, hence it is green. Thus $\mathbb{P}(X = 2, Y = 0) = \frac{1}{3} \cdot \frac{2}{5} = 2/15$ (initially the fraction of green is $1/3$; after removing a green, 5 balls remain, of which 2 are red; hence red on the second draw has probability $2/5$). The case $\{X = 2, Y = 1\}$ is analogous (first white, then red).

The event $\{X = 3, Y = 0\}$ means: red on the third draw with no white before it, thus two greens first. Its probability is $1/3 \cdot 1/5 \cdot 1/2 = 1/30$. Similarly, $\mathbb{P}(X = 3, Y = 2) =$

$1/3 \cdot 1/5 \cdot 1/2 = 1/30$, just with white in place of green.

If $X = 3$ and $Y = 1$, then the order is either white–green–red or green–white–red. Each has probability $1/3 \cdot 2/5 \cdot 2/4 = 1/15$, so together $2/15$.

If $X = 4$ and $Y = 1$, we have drawn the two greens and one white in some order, then red: patterns `zzfp`, `zfzp`, or `fzzp`. The probability is $1/3 \cdot 1/5 \cdot 1/2 \cdot 2/3 + 1/3 \cdot 2/5 \cdot 1/4 \cdot 2/3 + 1/3 \cdot 2/5 \cdot 1/4 \cdot 2/3 = 1/15$. Similarly, $\mathbb{P}(X = 4, Y = 2) = 1/15$ (swap the roles of white and green).

Finally, $\mathbb{P}(X = 5, Y = 2) = 1 -$ *(the sum of all other probabilities)*, but it can also be computed directly in the same manner.

For independence, every entry of the joint distribution must factor as the product of the corresponding marginals. This clearly fails for entries equal to 0. Consider $\mathbb{P}(X = 1, Y = 1) = 0$. Compute $\mathbb{P}(X = 1)$ and $\mathbb{P}(Y = 1)$ by summing the appropriate column/row:

$$\mathbb{P}(X = 1) = \mathbb{P}(X = 1, Y = 0) + \mathbb{P}(X = 1, Y = 1) + \mathbb{P}(X = 1, Y = 2) = 1/3 + 0 + 0 = 1/3,$$

$$\mathbb{P}(Y = 1) = \mathbb{P}(X = 1, Y = 1) + \mathbb{P}(X = 2, Y = 1) + \mathbb{P}(X = 3, Y = 1) + \mathbb{P}(X = 4, Y = 1) +$$
$$+ \mathbb{P}(X = 5, Y = 1) = 0 + 2/15 + 2/15 + 1/15 + 0 = 1/3.$$

Hence

$$0 = \mathbb{P}(X = 1, Y = 1) \neq \mathbb{P}(X = 1)\mathbb{P}(Y = 1) = \tfrac{1}{3} \cdot \tfrac{1}{3} = \tfrac{1}{9}.$$

4. a) Since X and Y are independent and each has expectation $1/p = 3/2$, we get $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y) = \frac{3}{2} \cdot \frac{3}{2} = \frac{9}{4} = 2.25$.

   b) $\mathbb{P}(X = 2 \mid Y = 5) = \mathbb{P}(X = 2) = (2/3) \cdot (1/3) = \dfrac{2}{9}$, by the equivalent definition of independence for discrete variables (and since $\mathbb{P}(Y = 5) > 0$).

   c*)
   $$\mathbb{P}(X = Y) = \mathbb{P}\Big(\bigcup_{k=1}^{\infty} \{X = Y = k\}\Big) = \sum_{k=1}^{\infty} \mathbb{P}(X = Y = k).$$
   By independence, $\mathbb{P}(X = Y = k) = \mathbb{P}(X = k)\mathbb{P}(Y = k)$ for every k, and thus using the geometric-series sum:
   $$\mathbb{P}(X = Y) = \sum_{k=1}^{\infty} p^2(1-p)^{2k-2} = \frac{p^2}{(1-p)^2}\sum_{k=1}^{\infty}(1-p)^{2k-2} = p^2 \sum_{l=0}^{\infty}((1-p)^2)^l$$
   $$= \frac{p^2}{1-(1-p)^2} \overset{p=2/3}{=} \frac{4}{9} \cdot \frac{1}{1-1/9} = \frac{1}{2}.$$

5. For the events $\{X = 2\}$ and $\{Y = 0\}$ to be independent, we need $\mathbb{P}(X = 2, Y = 0) = \mathbb{P}(X = 2)\mathbb{P}(Y = 0)$. Let the missing top-right entry be x and the bottom-left be y. Then $\mathbb{P}(X = 2) = x + \frac{1}{5}$, $\mathbb{P}(Y = 0) = \frac{1}{10} + \frac{1}{5} + x = \frac{3}{10} + x$, and $\mathbb{P}(X = 2, Y = 0) = x$. Thus
   $$\Big(x + \frac{1}{5}\Big)\Big(x + \frac{3}{10}\Big) = x \quad \Longleftrightarrow \quad x^2 - 0.5x + 0.06 = 0.$$

Solving gives $x_{1,2} = \dfrac{0.5 \pm \sqrt{0.5^2 - 4 \cdot 1 \cdot 0.06}}{2} = \dfrac{0.5 \pm 0.1}{2}$. The root $x_1 = 0.3$ is invalid, since with $x = 0.3$ the table sums to more than 1 (we do not yet know y, but it must be nonnegative, and the other entries already sum to 1.05). Hence $x = x_2 = 0.2 = 1/5$ is the only valid solution, and to make the table sum to 1 we need $y = 1/20$. The completed table is:

Y \ X	0	1	2
0	1/10	1/5	1/5
2	1/20	1/4	1/5

Using the definition of conditional probability and that $\mathbb{P}(X = k)$ equals the sum of the k-th column,

$$\mathbb{P}(XY > 0 \mid X < 2) = \frac{\mathbb{P}\Big((\{X = 1, Y = 2\} \cup \{X = Y = 2\}) \cap \{X < 2\}\Big)}{\mathbb{P}(X < 2)} =$$

$$= \frac{\mathbb{P}(X = 1, Y = 2)}{\mathbb{P}(X = 1) + \mathbb{P}(X = 0)} = \frac{\frac{1}{4}}{\frac{1}{10} + \frac{1}{20} + \frac{1}{5} + \frac{1}{4}} = \frac{5}{12}.$$

By linearity of expectation (omitting 0 terms) and using row sums for $\mathbb{P}(Y = l)$,

$$\mathbb{E}(4X + Y) = 4\mathbb{E}(X) + \mathbb{E}(Y) = 4\Big(\mathbb{P}(X = 1) + 2\mathbb{P}(X = 2)\Big) + 2\mathbb{P}(Y = 2) = \frac{9}{5} + \frac{16}{5} + 1 = 6.$$

Moreover, $XY \neq 0$ precisely when $(X, Y) = (1, 2)$ (then $XY = 2$) or $(X, Y) = (2, 2)$ (then $XY = 4$). Hence

$$\mathbb{E}(XY) = 2\,\mathbb{P}(X = 1, Y = 2) + 4\,\mathbb{P}(X = Y = 2) = 2 \cdot \frac{1}{4} + 4 \cdot \frac{1}{5} = \frac{13}{10}.$$

X and Y are not independent. For instance, from the above $\mathbb{E}(X) = 5/4$ and $\mathbb{E}(Y) = 1$, yet $\mathbb{E}(XY) = 13/10 \neq \mathbb{E}(X)\mathbb{E}(Y) = 5/4$. Also, one can find (k, l) with $\mathbb{P}(X = k, Y = l) \neq \mathbb{P}(X = k)\mathbb{P}(Y = l)$, e.g., for $k = l = 0$: $1/10 \neq (1/2) \cdot (3/10)$.

6. a) Yes. It is (weakly) increasing with limits 0 as $x \to -\infty$ and 1 as $x \to +\infty$. Left-continuity only needs to be checked at $x = 0$, since a constant function is continuous everywhere. As $\lim_{x \uparrow 0} F(x) = \lim_{x \uparrow 0} 0 = 0 = F(0)$, left-continuity holds at 0 (even though F is not continuous there). Hence F satisfies the conditions to be a distribution function.

   Note: F is the cdf of the a.s. constant 0 random variable. It is called the (left-continuous version of the) Heaviside function.

   b) Yes. The function is not only left-continuous but continuous (being a composition of continuous functions). Limits:

$$\lim_{x \to -\infty} F(x) = \lim_{x \to -\infty} e^{-e^{-x}} = e^{-\infty} = 0, \qquad \lim_{x \to \infty} F(x) = \lim_{x \to \infty} e^{-e^{-x}} = e^0 = 1.$$

   It is strictly increasing since its derivative is positive everywhere (chain rule):

$$F'(x) = -e^{-e^{-x}} \cdot (-e^{-x}) = e^{-e^{-x}} e^{-x} > 0.$$

Note: this is the cdf of the (type I) extreme value (Gumbel) distribution.

c) Not a distribution function, since the limits at $-\infty$ and $+\infty$ are the same (namely, 1).

d) Yes. We have $F'(x) = \frac{1}{1+x^2} > 0$, so F is strictly increasing; being differentiable everywhere, it is continuous (hence left-continuous). The limits are $\lim_{x\to-\infty} F(x) = \frac{1}{\pi}\cdot\left(-\frac{\pi}{2}\right) + \frac{1}{2} = 0$ and $\lim_{x\to\infty} F(x) = \frac{1}{\pi}\cdot\frac{\pi}{2} + \frac{1}{2} = 1$.

7. *For (*e) only final answers.*

For f to be a density, we need $f \geq 0$ (automatic in the given cases) and $\int_{-\infty}^{\infty} f = 1$. Where $f \equiv 0$, the integral contributes nothing. The cdf is given by $F(x) = \int_{-\infty}^{x} f(t)\,dt$; it suffices to start integrating from the smallest point where f is nonzero (below that $F(x) = 0$), and beyond the largest point after which $f \equiv 0$ we have $F(x) = 1$ (assuming f is indeed a density). The antiderivatives coincide with those used to determine α (with the correct α substituted). Since F is continuous at breakpoints, we may assign the endpoints to either adjacent interval.

(a)

$$1 \overset{!}{=} \int_0^2 \alpha(2x - x^2)\,dx = \alpha\Big[x^2 - x^3/3\Big]_0^2 = \alpha\,(4 - 8/3) = \tfrac{4}{3}\alpha,$$

so $\alpha = \dfrac{3}{4}$, and

$$F(x) = \begin{cases} 0, & x \leq 0, \\ \displaystyle\int_0^x \tfrac{3}{4}(2t - t^2)\,dt = \tfrac{3}{4}x^2 - \tfrac{1}{4}x^3, & 0 < x \leq 2, \\ 1, & 2 < x. \end{cases}$$

(b)

$$1 \overset{!}{=} \int_2^3 \alpha\sqrt{x-2}\,dx = \alpha\left[\tfrac{2}{3}(x-2)^{3/2}\right]_2^3 = \tfrac{2}{3}\alpha,$$

thus $\alpha = \dfrac{3}{2}$, and

$$F(x) = \begin{cases} 0, & x \leq 2, \\ \displaystyle\int_2^x \tfrac{3}{2}\sqrt{t-2}\,dt = (x-2)^{3/2}, & 2 < x \leq 3, \\ 1, & 3 < x. \end{cases}$$

(c) Analogous to (b), with different bounds:

$$1 \overset{!}{=} \int_3^4 \alpha\sqrt{x-2}\,dx = \tfrac{2}{3}\alpha\Big((4-2)^{3/2} - (3-2)^{3/2}\Big) = \tfrac{2}{3}\alpha(2\sqrt{2} - 1),$$

hence $\alpha = \dfrac{3}{2(2\sqrt{2}-1)}$, and

$$F(x) = \begin{cases} 0, & x \leq 3, \\ \displaystyle\int_3^x \frac{3}{2(2\sqrt{2}-1)}\sqrt{t-2}\,dt = \frac{(x-2)^{3/2} - 1}{2\sqrt{2}-1}, & 3 < x \leq 4, \\ 1, & 4 < x. \end{cases}$$

(d)

$$1 \overset{!}{=} \int_0^{\pi} \alpha \cos(x/2)\, dx = \Big[2\alpha \sin(x/2)\Big]_0^{\pi} = 2\alpha,$$

so $\alpha = \dfrac{1}{2}$, and

$$F(x) = \begin{cases} 0, & x \leq 0, \\ \displaystyle\int_0^x \tfrac{1}{2}\cos(t/2)\, dt = \sin\!\left(\tfrac{x}{2}\right), & 0 < x \leq \pi, \\ 1, & \pi < x. \end{cases}$$

(*e) a) 1 b) 2.63 c) 3.54 — obtainable e.g. by numerical root-finding (in (a) the cubic can also be solved symbolically, and by inspection $x = 1$ is the unique valid root). d) $\dfrac{\pi}{3}$ — the unique angle in $(0, \pi)$ whose half-angle sine equals 1/2.

8. Y has cdf

$$F_Y(y) = \mathbb{P}(Y < y) = \mathbb{P}(3X - 1 < y) = \mathbb{P}\Big(X < \frac{y+1}{3}\Big).$$

Since X is integer-valued, we round $\frac{y+1}{3}$ to an integer threshold. For $y = \pi$,

$$F_Y(\pi) = \mathbb{P}\Big(X \leq \Big\lfloor \tfrac{\pi+1}{3} \Big\rfloor\Big) = \mathbb{P}(X \leq 1).$$

As $X \sim \text{Poisson}(3)$,

$$\mathbb{P}(X \leq 1) = \frac{3^0 e^{-3}}{0!} + \frac{3^1 e^{-3}}{1!} = e^{-3} + 3e^{-3} = 4e^{-3},$$

so $F_Y(\pi) = 4e^{-3} \approx 0.1991$.

9. Both F_X and F_Y are step functions. F_X has jumps at the possible die outcomes, each jump of size 1/6. Crossing such a point increases the probability of being below that value by 1/6. Y takes values in $\{1, 2, 3, 4\}$ (the number of divisors of the rolled face), so F_Y jumps at these points.
The divisor counts are: $Y(1) = 1$, $Y(2) = Y(3) = Y(5) = 2$, $Y(4) = 3$, $Y(6) = 4$.

$$F_X(x) = \begin{cases} 0, & x \leq 1, \\ 1/6, & 1 < x \leq 2, \\ 1/3, & 2 < x \leq 3, \\ 1/2, & 3 < x \leq 4, \\ 2/3, & 4 < x \leq 5, \\ 5/6, & 5 < x \leq 6, \\ 1, & 6 < x, \end{cases} \qquad F_Y(y) = \begin{cases} 0, & y \leq 1, \\ 1/6, & 1 < y \leq 2, \\ 2/3, & 2 < y \leq 3, \\ 5/6, & 3 < y \leq 4, \\ 1, & 4 < y. \end{cases}$$

They are not continuous.

10. The condition implies $d < 9$, since for any $a < d$ we have $\mathbb{P}(S > a) = 1$.

a) For any interval, its probability is the integral of the density over that interval. Thus

$$\mathbb{P}(S>9)=\int_9^\infty f_S(x)\,dx=\int_9^\infty c\,x^{-5/2}\,dx=c\Big[-\tfrac{2}{3}x^{-3/2}\Big]_9^\infty=c\cdot\tfrac{2}{3}\cdot 9^{-3/2}=c\cdot\tfrac{2}{3}\cdot\tfrac{1}{27}.$$

Given $\mathbb{P}(S>9)=0.2689$, we obtain $c=10.89$.

b) Any density integrates to 1 over $\mathbb{R}$. Using $f_S(x)=0$ for $x<d$,

$$1=\int_{-\infty}^\infty f_S(x)\,dx=\int_d^\infty c\,x^{-5/2}\,dx=c\Big[-\tfrac{2}{3}x^{-3/2}\Big]_d^\infty=c\cdot\tfrac{2}{3}\,d^{-3/2}.$$

Hence

$$d^{3/2}=c\cdot\tfrac{2}{3}\quad\Longrightarrow\quad d=\Big(c\cdot\tfrac{2}{3}\Big)^{2/3}=3.749.$$

11. (a) For $\omega\in\Omega=[0,1]$, the distance from 0.3 is $|\omega-0.3|$, so

$$X(\omega)=|\omega-0.3|=\begin{cases}0.3-\omega, & 0\le\omega\le 0.3,\\ \omega-0.3, & 0.3\le\omega\le 1.\end{cases}$$

(b) Since a point in $[0,1]$ is at most 0.7 away from 0.3, we have $\mathrm{Ran}(X)=[0,0.7]$, $F_X(x)=0$ for $x<0$ and $F_X(x)=1$ for $x>0.7$.
For $0<x<0.3$, there are points at distance $<x$ on both sides of 0.3:

$$\mathbb{P}(X<x)=\mathbb{P}([0.3-x,0.3])+\mathbb{P}([0.3,0.3+x])=x+x=2x,$$

using geometric probability in one dimension (probability proportional to interval length, total length 1). For $0.3\le x\le 0.7$,

$$\mathbb{P}(X<x)=\mathbb{P}([0,0.3])+\mathbb{P}([0.3,0.3+x])=0.3+x.$$

Thus

$$F_X(x)=\begin{cases}0, & x\le 0,\\ 2x, & 0<x\le 0.3,\\ 0.3+x, & 0.3<x\le 0.7,\\ 1, & 0.7<x.\end{cases}$$

(Endpoints may be assigned arbitrarily to adjacent pieces since the cdf is continuous.)
(c) Differentiating (and setting the density to 0 where F_X is not differentiable),

$$f_X(x)=\begin{cases}2, & 0<x<0.3,\\ 1, & 0.3<x<0.7,\\ 0, & \text{otherwise.}\end{cases}$$

Remark: Although the intuitive "slope" at $x=0.3$ equals $\frac{3}{2}$, it is acceptable to set $f_X(0.3)$ arbitrarily (e.g. 0), since F_X is not differentiable there; modifying f_X at a single point does not change $F_X(x)=\int_{-\infty}^x f_X(t)\,dt$.

12. (a) For $P = \omega \in [0,1]$,

$$X(\omega) = \min\{\omega^2, (1-\omega)^2\} = \begin{cases} \omega^2, & 0 \le \omega \le \frac{1}{2}, \\ (1-\omega)^2, & \frac{1}{2} \le \omega \le 1. \end{cases}$$

(b) Then $\mathrm{Ran}(X) = [0, 1/4]$. Hence $F_X(x) = 0$ for $x \le 0$ (also at $x = 0$ by continuity) and $F_X(x) = 1$ for $x \ge 1/4$. For $0 < x < 1/4$,

$$\begin{aligned} F_X(x) &= \mathbb{P}(X < x) = \mathbb{P}(\{\omega \in [0,1] : X(\omega) < x\}) \\ &= \mathbb{P}(\{\omega \in [0,1/2] : \omega^2 < x\}) + \mathbb{P}(\{\omega \in (1/2,1] : (1-\omega)^2 < x\}) \\ &= \mathbb{P}([0, \sqrt{x})) + \mathbb{P}((1-\sqrt{x}, 1]) = 2\sqrt{x}. \end{aligned}$$

Thus

$$F_X(x) = \begin{cases} 0, & x \le 0, \\ 2\sqrt{x}, & 0 < x \le \frac{1}{4}, \\ 1, & \frac{1}{4} < x, \end{cases} \qquad f_X(x) = \begin{cases} \dfrac{1}{\sqrt{x}}, & 0 < x < \frac{1}{4}, \\ 0, & \text{otherwise.} \end{cases}$$

13. a) The given triangle (call it Ω) consists exactly of those points in $(0,1)^2$ whose y-coordinate exceeds the x-coordinate:

$$\Omega = \{(x,y) \in (0,1) \times (0,1) : x < y\}.$$

Its area is $T_\Omega = 1/2$. For $x \in (0,1)$, to find $F_X(x) = \mathbb{P}(X < x)$ it is simpler to compute $\mathbb{P}(X \ge x)$: the event $\{X \ge x\}$ occurs if (X,Y) falls into the right isosceles triangle with vertices (x,x), $(x,1)$, $(1,1)$, whose area is $\frac{(1-x)^2}{2}$. Therefore,

$$\mathbb{P}(X < x) = 1 - \mathbb{P}(X \ge x) = 1 - \frac{(1-x)^2/2}{1/2} = 2x - x^2.$$

Since $X \in [0,1]$ a.s., we obtain

$$F_X(x) = \begin{cases} 0, & x \le 0, \\ 2x - x^2, & 0 < x \le 1, \\ 1, & x > 1, \end{cases} \qquad f_X(x) = \begin{cases} 2 - 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

b) Similarly, for $y \in (0,1)$ the event $\{Y < y\}$ is that (X,Y) lies in the right triangle with vertices $(0,0)$, $(0,y)$, (y,y), with area $y^2/2$. Dividing by T_Ω yields

$$F_Y(y) = \begin{cases} 0, & y \le 0, \\ y^2, & 0 < y \le 1, \\ 1, & y > 1, \end{cases} \qquad f_Y(y) = \begin{cases} 2y, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

14. a) For $\omega = (x, y) \in \Omega$, the distance to the x-axis is the second coordinate, so $X(x, y) = y$.
b) $F_X(t) = \mathbb{P}(X < t)$ is the probability that P lies within distance t of the x-axis. Within Ω, all distances to the x-axis lie in $[0, 1/2]$. For $0 < t \le \frac{1}{2}$, the favorable set inside the triangle is a trapezoid. Its area equals the total area minus the area of the top triangle:

$$\frac{1}{4} - \left(\frac{1}{2} - t\right)^2 = \frac{1}{4} - \left(\frac{1}{4} - t + t^2\right) = t - t^2 + \frac{1}{4} - \frac{1}{4} = t - t^2.$$

Dividing by the total area $\frac{1}{4}$ gives

$$\mathbb{P}(X < t) = \frac{\frac{1}{4} - \left(\frac{1}{2} - t\right)^2}{\frac{1}{4}} = 4t - 4t^2.$$

Hence

$$F_X(t) = \begin{cases} 0, & t \le 0, \\ 4t - 4t^2, & 0 < t \le \frac{1}{2}, \\ 1, & \frac{1}{2} < t, \end{cases} \qquad f_X(t) = \begin{cases} 4 - 8t, & 0 < t < \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

15. (a) Let the unit square be $[0, 1]^2$. We may assume the random point on the bottom edge is $(a, 0)$ with $a \sim \text{Unif}[0, 1]$, and the one on the top edge is $(b, 1)$ with $b \sim \text{Unif}[0, 1]$, independent of a.
Denote by D the distance between a and b. By Pythagoras, $X = D^2 = (b - a)^2 + 1$ (see Fig. 1).
View the experiment as choosing $(a, b) \in [0, 1]^2$ uniformly (Fig. 2). Since $|b - a| \le 1$, we have $\text{Ran}(X) = [1, 2]$, so $F_X(x) = 0$ for $x \le 1$ and $F_X(x) = 1$ for $x > 2$. For $1 < x \le 2$,

$$\begin{aligned} F_X(x) &= \mathbb{P}(X < x) = \mathbb{P}((b - a)^2 + 1 < x) = \mathbb{P}(|b - a| < \sqrt{x - 1}) \\ &= \mathbb{P}(0 < b - a < \sqrt{x - 1}) + \mathbb{P}(0 < a - b < \sqrt{x - 1}). \end{aligned}$$

The unfavorable area consists of two right isosceles triangles each of area $(1 - \sqrt{x - 1})^2/2 = (1 - 2\sqrt{x - 1} + x - 1)/2 = -\sqrt{x - 1} + x/2$. Hence the favorable area (and thus the probability, since the total area is 1) is

$$F_X(x) = 1 + 2\sqrt{x - 1} - x.$$

In summary,

$$F_X(x) = \begin{cases} 0, & x \le 1, \\ 1 + 2\sqrt{x - 1} - x, & 1 < x \le 2, \\ 1, & 2 < x, \end{cases} \qquad f_X(x) = \begin{cases} \dfrac{1}{\sqrt{x - 1}} - 1, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(c) The density is strictly decreasing in x, hence it is maximal near 1 (indeed tends to ∞ as $x \downarrow 1$).

16. First, $\mathrm{Ran}(Y) = (0,1)$, hence $F_Y(x) = \mathbb{P}(Y < x) = 0$ for $x \le 0$ or $x \ge 1$, and $f_Y(x) = 0$ for $x \le 0$ or $x \ge 1$.

Let the three random points be X_1, X_2, X_3. The probability that any two coincide is 0, so we ignore that event and assume the three are distinct.

For $x \in (0,1)$, $Y < x$ (up to a null event) iff at least two of X_1, X_2, X_3 are less than x. Any two of the events $\{X_1 < x, X_2 < x\}$, $\{X_1 < x, X_3 < x\}$, $\{X_2 < x, X_3 < x\}$ intersect in $\{X_1 < X_2 < X_3 < x\}$, and the triple intersection is the same. By the inclusion–exclusion principle,

$$\begin{aligned}
F_Y(x) &= \mathbb{P}(Y < x) \\
&= \mathbb{P}(X_1 < x, X_2 < x) + \mathbb{P}(X_1 < x, X_3 < x) + \mathbb{P}(X_2 < x, X_3 < x) \\
&\quad - 2\,\mathbb{P}(X_1 < x, X_2 < x, X_3 < x) \\
&= 3x^2 - 2x^3,
\end{aligned}$$

using independence and uniformity (or a 3-dimensional geometric probability argument). Therefore

$$F_Y(x) = \begin{cases} 0, & x \le 0, \\ 3x^2 - 2x^3, & 0 < x \le 1, \\ 1, & 1 < x, \end{cases} \qquad f_Y(x) = \begin{cases} 6x - 6x^2, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$